[S26] DATABASES COURSE

ASSIGNMENT 1: ER DIAGRAM

REPORT

TEAM 37

February, 2026

INNOPOLIS UNIVERSITY

CONTEXT INFO

Team members

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Note: Through all time of working on this assignment **the whole team** was discussing raised problems and helping each other.

Team meetings

First meeting	18.02.2026	All people from the team gathered, made a group chat, read the assignment file and planned the next meeting.
Second meeting	23.02.2026	All members met to define the roles, discuss the assignment and start the work together.
Third meeting	28.02.2026	Final meeting to review the ER diagram, PostgreSQL script and report. Last fixes and submission to Moodle. Also lab presentation was discussed and practiced.

P.S. The design and features taken from lecture slides were approved for usage by Marko Pezer personally.

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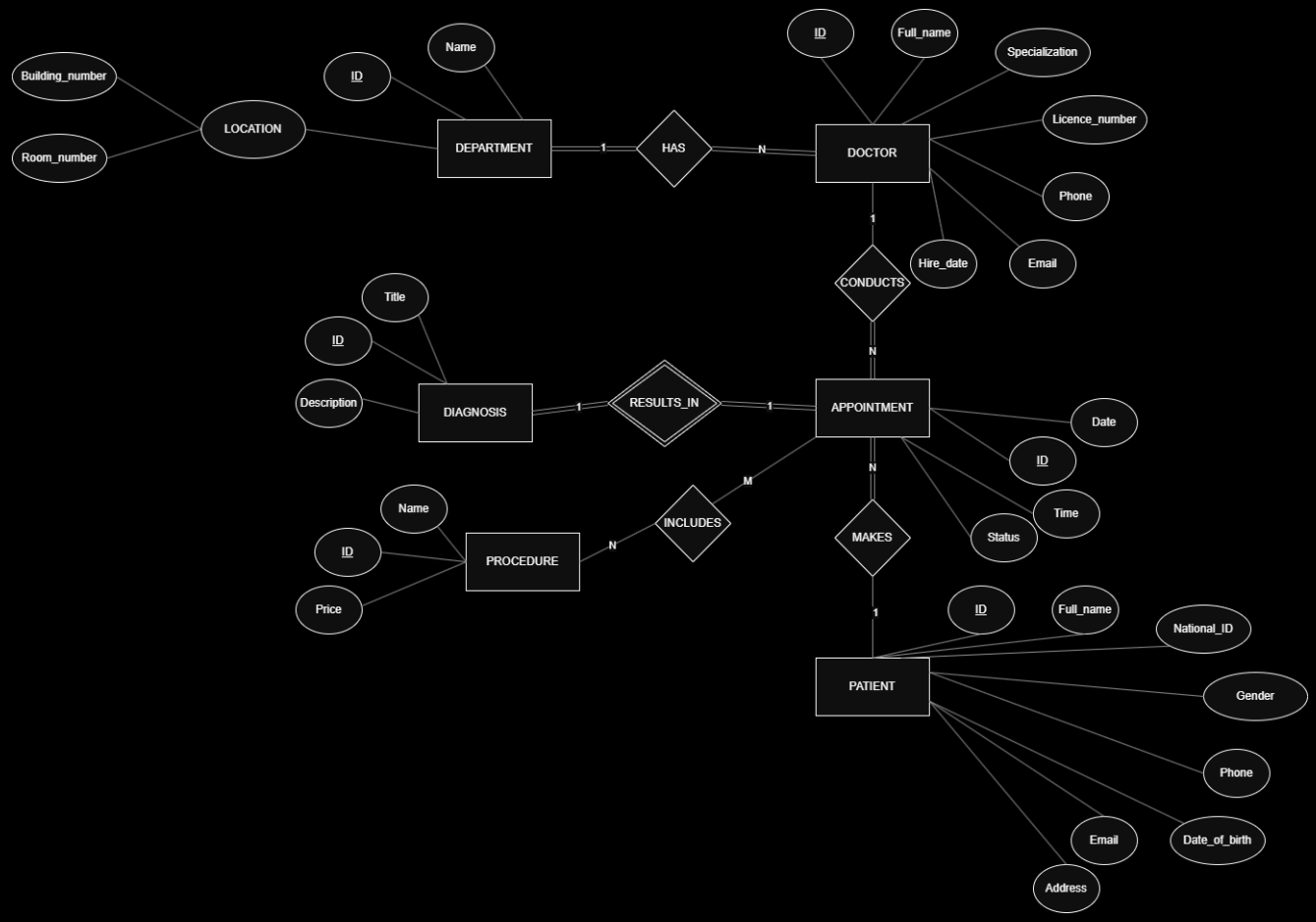
CORE CONCEPT

ER Diagram



Important Design Decisions

- The LOCATION (composite attribute) was created to avoid duplication and conflicts between DEPARTMENTS.
- DEPARTMENT and DOCTOR have total participation both, meaning that every department must have at least one doctor, and every doctor must be assigned to some department.
- DIAGNOSIS was kept as a strong entity with its own ID, but made fully dependent on APPOINTMENT through a mandatory unique foreign key.
- The relation M:N for PROCEDURE and APPOINTMENT was simplified to Chen ER notation. While in PostgreSQL file it is implemented as an associative entity.
- LOCATION is modeled as a composite attribute of the DEPARTMENT entity.

Assumptions about Requirements

- It was assumed that each department has exactly one location, defined only by building and room number.

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- Also it was assumed that phone numbers and emails must be unique for every doctor and patient.
- At the beginning of the work the team decided that the DIAGNOSIS can be uniquely identified, so then it was implemented further like a strong entity.
- Assumed that each department exists in only one physical location. Location is just a property of a department.

Usage of AI tools

AI was used to generate random data to fulfil the tables in PostgreSQL script, to check any grammar mistakes in report document.

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